Piper PA-46-310P Malibu (MALIBU)

Continental TSIO-520-BE, twin-turbocharged and intercooled, 310 hp · 6 seats · Typical PA-46-310P (1984–88, TSIO-520-BE, 4,100 lb, 5.5 psi cabin, hydraulic gear) POH procedures; analogue panel with King KFC 150 autopilot. Deice equipment and standby alternator where fitted; TIT and manifold pressure limits vary slightly with POH revision – use the placarded red lines.

V-speeds & limits

Vr	**70–74 KIAS (liftoff)**
Vx	**81 KIAS**
Vy	**110 KIAS**
Best glide	**90 KIAS (gear & flaps up, prop low RPM)**
Vfe	**165 KIAS (10°) / 130 KIAS (20°) / 116 KIAS (36° full)**
Vlo	**165 KIAS extend / 130 KIAS retract**
Vle	**165 KIAS**
Va	**135 KIAS (4,100 lb)**
Vno	**173 KIAS**
Vne	**203 KIAS**
Vs / Vso	**approx 69 / 58 KIAS (max weight)**
Vref	**80 KIAS final (full flaps); approx 75 KIAS short field**
Emergency descent	**165 KIAS gear down / 135 KIAS in turbulence**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel sumps, deice boots and static wicks, tyres, cowl and turbo area, no oil leaks)**
Weight & balance / takeoff data	**COMPUTED (4,100 lb max takeoff, 4,118 lb ramp)**
Passenger briefing	**COMPLETE (belts, airstair door, emergency exit, oxygen masks, no smoking)**
Seats, belts, shoulder harnesses	**ADJUST & LOCK**
Cabin door (upper & lower) / baggage doors	**CLOSED & LATCHED (DOOR AJAR annunciator out once power is on)**
Parking brake	**SET**
Landing gear selector	**DOWN**
Emergency gear extension lever	**NORMAL / STOWED (floor, between front seats)**
Throttle / propeller / mixture	**CLOSED / FULL FORWARD / IDLE CUTOFF**
Fuel selector	**FULLEST TANK (LEFT or RIGHT – OFF is detented, there is no BOTH)**
Fuel quantity	**CHECK both gauges (120 gal usable total)**
Avionics master	**OFF**
Battery master	**ON – voltage CHECK (approx 24 V)**
Annunciator panel	**PRESS TO TEST (all lights; gear lights three green)**
Cabin pressurisation dump switch	**NORMAL (not DUMP)**
Pressurisation controller	**SET field elevation + 500 ft; rate knob approx mid-range**
Oxygen	**CHECK pressure (approx 1,800 psi full) and quantity for flight; masks available**
Alternate static source	**OFF (normal)**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR**
Mixture	**FULL RICH**
Propeller	**FULL FORWARD (high RPM)**
Throttle	**FULL OPEN**
Fuel pump	**HI until fuel flow peaks (approx 3–5 sec), then OFF**
Throttle	**APPROX 1/4 OPEN**

▶ Engine hot or flooded?

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Starter	**ENGAGE (release when engine fires; 30 sec max, then cool)**
Throttle	**1,000 RPM**
Oil pressure	**10 psi within 30 sec (normal 30–60 psi)**
Fuel pump	**OFF (LO if fuel flow fluctuates)**
Alternator switch	**ON – charging (ammeter / voltage approx 28 V)**
Standby alternator	**ON / ARMED (if fitted – STBY ALT light out)**
Avionics master	**ON**
Engine instruments	**CHECK (oil pressure, oil temp rising, fuel flow, TIT, CHT)**

Before Taxi

Brakes	**CHECK on initial roll**
Flight instruments	**CHECK & SET (altimeter, HSI slaved, attitude erect, turn coordinator in turns)**
Avionics / GPS	**SET**
KFC 150 autopilot / electric trim	**PREFLIGHT TEST – disconnect checks, trim both directions – OFF**
Pitot heat / stall warning heat	**ON momentarily – CHECK (ammeter rise), then AS REQUIRED**
Propeller deice	**ON – ammeter in green, then AS REQUIRED (if fitted)**
Windshield hot plate	**ON – CHECK, then AS REQUIRED (if fitted)**
Surface deice (boots)	**SINGLE CYCLE – pneumatic pressure in green, boots inflate / deflate (if fitted)**
Elevator / rudder trim	**SET for takeoff**
Fuel selector	**CHECK fullest tank (LEFT or RIGHT)**
Flaps	**SET for takeoff (0° normal; 10° short field) – indicator CHECK**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Mixture	**FULL RICH**
Throttle	**1,700 RPM**
Magnetos	**CHECK L & R (150 RPM max drop, 50 RPM max difference)**
Propeller	**CYCLE 2–3 times (RPM drops, then returns)**
Engine instruments	**CHECK (oil pressure 30–60 psi, oil temp in green, fuel flow steady, TIT / CHT)**
Alternator / standby alternator	**CHECK charging, voltage approx 28 V**
Suction / pneumatic pressure	**CHECK in green (gyro suction approx 4.5–5.2 in Hg; deice pressure in green)**
Throttle	**IDLE check, then 1,000 RPM**
Fuel selector	**FULLEST TANK (feel for detent)**
Fuel pump	**LO**
Pressurisation	**Dump switch NORMAL; controller field elevation + 500 ft; door seal inflated, DOOR AJAR out**
Oxygen	**ON / AVAILABLE as required**
Propeller	**FULL FORWARD**
Flaps	**SET & INDICATING (0° or 10°)**
Trim	**SET takeoff range**
Landing gear selector	**DOWN – three green**
Ice protection / pitot heat	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Doors & windows	**LATCHED (DOOR AJAR out)**
Takeoff briefing	**COMPLETE (abort point; throttle slowly to red line – do not exceed max manifold pressure; gear up with positive rate and no usable runway, below 130 KIAS)**

After Takeoff / Climb

Takeoff power	**FULL THROTTLE applied slowly – manifold pressure at or below red line (approx 36–38 in Hg per placard), 2,600 RPM**
Landing gear	**UP (positive rate, no usable runway, below 130 KIAS) – lights out, HYD PUMP light out**
Flaps	**UP (above approx 85 KIAS, obstacles cleared)**
Climb power	**REDUCE SLOWLY to approx 32 in Hg / 2,500 RPM (cruise climb)**
Mixture	**FULL RICH / climb fuel flow per POH schedule (TIT below 1,650°F)**
Fuel pump	**OFF above 1,000 ft AGL (LO if fuel flow fluctuates)**
Airspeed	**110 KIAS (Vy) / 120–130 KIAS cruise climb for cooling**
Pressurisation	**CHECK cabin climbing on rate, differential increasing (5.5 psi max); controller SET cruise altitude + 500 ft**
Engine instruments	**CHECK (CHT, TIT, oil temp 240°F max, oil pressure)**
Ice protection	**AS REQUIRED**
Landing / taxi lights	**OFF**

Cruise

Cruise power	**SET (2,300–2,400 RPM, approx 26–31 in Hg; 75% max recommended) – change power slowly**
Mixture	**LEAN SLOWLY per POH cruise table – TIT 1,650°F max recommended cruise (1,750°F red line)**
Fuel selector	**ALTERNATE LEFT / RIGHT to keep tanks balanced – note time**
Fuel pump	**LO if fuel flow fluctuates (vapour suppression at altitude / hot fuel), otherwise OFF**
Fuel quantity / flow	**CHECK against plan**
Pressurisation	**CHECK cabin altitude, rate zero, differential 5.5 psi max (cabin approx 8,000 ft at FL250)**
Oxygen	**CHECK pressure; masks available (CABIN ALT warning approx 10,000 ft cabin)**
Engine instruments	**CHECK (oil 30–60 psi / 240°F max, CHT, TIT)**
Ice protection	**AS REQUIRED**
Altimeter / heading	**CHECK**

Descent

Altimeter	**SET**
Pressurisation controller	**SET destination field elevation + 500 ft; rate approx 500 fpm (cabin ahead of aircraft)**
Power	**REDUCE SLOWLY (approx 1 in Hg per minute) – keep approx 18–20 in Hg minimum to hold cabin pressure**
Mixture	**ENRICH progressively (TIT / CHT monitored)**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Windshield heat / ice protection	**AS REQUIRED**
Airspeed	**BELOW 173 KIAS (Vno); plan below 165 KIAS before gear extension**
Approach briefing	**COMPLETE (Vref 80, flap setting, go-around power applied slowly)**
Seat belts / harnesses	**SECURE**

Before Landing

Pressurisation	**CHECK cabin altitude at or above field elevation, differential approaching zero**
Fuel selector	**FULLEST TANK (LEFT or RIGHT)**
Fuel pump	**LO**
Mixture	**FULL RICH (as required at high field elevation)**
Airspeed	**BELOW 165 KIAS**
Landing gear	**DOWN – three green, HYD PUMP light out**
Flaps	**10° below 165 KIAS; 20° below 130 KIAS; 36° on final below 116 KIAS**
Propeller	**FULL FORWARD (on final)**
Autopilot / yaw damper	**OFF**
Airspeed	**80 KIAS final (full flaps; approx 75 KIAS short field; add 10–15 KIAS with residual ice)**
Landing light	**ON**
Landing gear	**THREE GREEN – CONFIRMED on final**

After Landing

Flaps	**UP**
Pressurisation	**CONFIRM depressurised (differential zero; dump if required)**
Fuel pump	**OFF**
Ice protection	**OFF (pitot heat, prop deice, windshield heat, boots)**
Landing light / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trim	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttle	**1,000 RPM – turbocharger cool-down 3–5 min (TIT stabilised / falling)**
Avionics master	**OFF**
Electrical equipment / lights	**OFF**
Oxygen	**OFF**
Mixture	**IDLE CUTOFF**
Magnetos	**OFF (after propeller stops)**
Alternator / standby alternator / battery master	**OFF**
Control lock	**INSTALL**
Chocks / tiedowns / covers	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Standby Alternator

ALT INOP annunciator, ammeter zero or voltage below approx 25 V

Alternator switch ..... **OFF, then ON (reset)**

Alternator field circuit breaker ..... **CHECK / RESET once**

▶ Alternator restored?

Yes:

- Ammeter / voltmeter ..... **MONITOR**

No – standby alternator fitted:

- Alternator switch ..... **OFF**
- Standby alternator ..... **ON – STBY ALT ON light; voltage CHECK**
- Electrical load ..... **REDUCE to within standby alternator capacity (approx 20 A: essential avionics, pitot heat; windshield heat and prop deice OFF)**
- Landing gear / flaps ..... **PLAN early gear extension – hydraulic pump and flaps draw heavily**
- Deferred: Main alternator failed – standby alternator, limited electrical load; land as soon as practical. ..... **NOTED**

No – battery only:

- Alternator switch ..... **OFF**
- Non-essential electrical load ..... **OFF (avionics to minimum, deice OFF, lights OFF)**
- Landing gear ..... **EXTEND EARLY while battery power remains (free-fall extension if the pump fails)**
- Pressurisation ..... **UNAFFECTED (pneumatic), but CABIN ALT warning and gear lights may fail – monitor**
- Deferred: Alternator failed – battery only; extend gear early, flaps-up landing possible, land as soon as practical. ..... **NOTED**

Low Oil Pressure

Oil pressure below 30 psi in flight

Oil pressure / oil temperature ..... **CHECK and cross-check**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

- Power ..... **REDUCE to minimum required (turbochargers are oil-lubricated – expect failure)**
- Airport ..... **DIVERT to nearest suitable, stay within glide of a landing area**
- Altitude ..... **HOLD altitude as long as practicable for glide range**
- Deferred: Low oil pressure – land as soon as possible, expect possible engine failure. ..... **NOTED**

No – gauge suspect:

- Flight ..... **CONTINUE, monitor oil temperature and pressure**
- Deferred: Oil pressure indication suspect – maintenance check required. ..... **NOTED**

Turbocharger Failure / Manifold Pressure Loss

Manifold pressure falls or cannot be obtained, fuel flow high for the MP, rough running, possible exhaust smell

Throttle ..... **REDUCE, then re-set smoothly**

Mixture ..... **LEAN to smooth running at the manifold pressure available (fuel flow too high for MP)**

Engine instruments ..... **CHECK (TIT, CHT, oil pressure and temperature)**

Exhaust leak indications ..... **CHECK (smell, smoke, discoloured cowl, cabin CO) – exhaust gas in cowling is a fire hazard**

▶ Exhaust leak or fire suspected?

Yes:

- Power ..... **MINIMUM required**
- Cabin heat ..... **OFF (CO risk)**
- Landing ..... **NEAREST SUITABLE AIRPORT – be ready for Engine Fire in Flight**
- Deferred: Turbocharger / exhaust failure – fire hazard, land as soon as possible, no go-around. ..... **NOTED**

No – power loss only:

- Altitude ..... **DESCEND to where naturally-aspirated power is adequate (approx 25 in Hg available near sea level, less with altitude)**
- Pressurisation ..... **EXPECT cabin to climb (turbo bleed air lost) – oxygen ON above 10,000 ft cabin, descend**
- Deferred: Turbocharger failed – reduced power, no pressurisation, longer takeoff distance; land as soon as practical. ..... **NOTED**

High TIT / Overboost

TIT above 1,750°F red line or manifold pressure above red line

▶ Indication?

Overboost (MP above red line):

- Throttle — **REDUCE below red line, re-apply slowly**
- Oil temperature — **CHECK (cold oil slows the controller – warm up before full power)**

Momentary overboost of approx 1 in Hg for a few seconds is tolerated; sustained or large overboost requires a maintenance inspection before further flight.

Deferred: Overboost occurred – record peak MP and duration, maintenance inspection required. — **NOTED**

High TIT:

- Mixture — **ENRICH until TIT below 1,650°F**
- Throttle — **REDUCE if TIT remains high**
- Fuel flow — **CHECK against POH for the power set (low flow – fuel pump LO)**

▶ TIT within limits?

Yes:

- Flight — **CONTINUE, richer mixture**

No:

- Power — **REDUCE to the highest setting that keeps TIT below 1,650°F**

Deferred: High TIT – reduced power, suspect fuel system or TIT probe; maintenance check. — **NOTED**

Fuel Flow Fluctuation / Low Fuel Pressure

- Fuel pump — **LO (vapour suppression); HI only if fuel flow is very low – may flood engine at low power**
- Fuel selector — **SWITCH TANKS (check quantity in the tank being used)**
- Mixture — **ADJUST for smooth running**
- Engine instruments — **CHECK (fuel flow, TIT, CHT)**

▶ Fuel flow steady and engine smooth?

Yes:

Deferred: Fuel flow fluctuation cleared – leave fuel pump LO; report to maintenance. — **NOTED**

No:

- Power — **REDUCE to smoothest setting; lower altitude helps vapour**

Deferred: Fuel flow unstable – land as soon as practical at nearest suitable airport. — **NOTED**

Pressurisation Controller Fault / Cabin Not Holding

Cabin altitude or rate not following the controller; differential abnormal

- Cabin pressurisation dump switch — **NORMAL**
- Pressurisation controller — **RESELECT cabin altitude / rate; cycle rate knob**
- Manifold pressure — **ABOVE approx 20 in Hg (bleed air inflow needs power)**
- Door seal / DOOR AJAR — **CHECK (hiss at door, light out)**
- Cabin differential — **CHECK – 5.5 psi max (safety valve relieves above)**

▶ Cabin response?

Cabin following again:

- Pressurisation — **MONITOR closely**

Deferred: Pressurisation controller intermittent – maintenance check. — **NOTED**

Cabin climbing:

- Oxygen — **ON if cabin above 10,000 ft**
- Altitude — **DESCEND to keep cabin below 10,000 ft; Rapid Decompression procedure if CABIN ALT warning**

Deferred: Pressurisation failed – unpressurised or limited altitude, oxygen as required. — **NOTED**

Differential too high:

- Altitude / power — **DESCEND; reduce power to reduce inflow**
- Before landing — **DUMP below 10,000 ft so the cabin is depressurised at touchdown**

Deferred: Outflow valve fault – dump before landing; do not open the door until differential is zero. — **NOTED**

Door Seal Failure / Door Open in Flight

DOOR AJAR light, loud hiss at door, cabin will not hold pressure

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 110–120 KIAS**

Belts / loose items ..... **SECURE; passengers away from the door**

▶ Door seal only (door latched) or door unlatched?

Seal failed, door latched:

Altitude ..... **DESCEND – cabin cannot hold pressure; oxygen if cabin above 10,000 ft**

Deferred: Door seal failed – unpressurised, remain at or below 10,000–12,500 ft, maintenance required. ..... **NOTED**

Door unlatched:

Cabin pressurisation ..... **DUMP (do not pressurise)**

Do not attempt to close or re-latch the door in flight. Expect noise and buffet.

Deferred: Cabin door unlatched – unpressurised, return and land as soon as practical, normal approach speed. ..... **NOTED**

Landing Gear Unsafe / Fails to Retract or Extend

Red gear warning light, fewer than three green, or HYD PUMP light stays on

Airspeed ..... **BELOW 165 KIAS (below 130 KIAS to retract)**

Gear indicator bulbs ..... **CHECK (press to test / dim switch) – navigation lights dim the gear lights**

Hydraulic pump circuit breaker ..... **CHECK / RESET once**

Landing gear selector ..... **CYCLE once to the desired position**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM three green / lights out**

Deferred: Gear system fault – do not retract after landing; hydraulic / gear inspection required. ..... **NOTED**

No – failed to retract:

Landing gear selector ..... **DOWN – leave down**

Hydraulic pump ..... **If HYD PUMP light stays on, PULL hydraulic pump circuit breaker (pump running against a leak)**

Airspeed ..... **BELOW 165 KIAS**

Deferred: Gear will not retract – gear down, 165 KIAS max, reduced range; land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Emergency Gear Extension (Free Fall).

Flap Failure / Asymmetry

Flap selector ..... **STOP at current position – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Landing ..... **AT CURRENT FLAP SETTING – approach approx 90 KIAS flaps up, longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, approach approx 90 KIAS, increased landing distance. ..... **NOTED**

Propeller Overspeed

RPM above 2,600 and not responding to the prop lever

Throttle ..... **REDUCE**

Propeller ..... **FULL AFT (low RPM)**

Airspeed ..... **REDUCE (pitch up) to bring RPM within limits**

Oil pressure ..... **CHECK (governor uses engine oil)**

Power ..... **SET minimum that keeps RPM at or below 2,600**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

Vacuum / Pneumatic Pressure Loss

Suction gauge low, VAC annunciator, or deice pressure low

Suction / pneumatic pressure gauges ..... **CHECK (dual pumps – one failed leaves reduced capacity)**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, electric standby instruments**

▶ **Suction and deice pressure?**

One pump failed – suction still in green:

- Surface deice ..... **USE SPARINGLY – single pump; exit icing conditions**
- Deferred: One air pump failed – gyros on remaining pump, limited boot capacity; avoid icing. ..... **NOTED**

Suction lost:

- Failed gyro instruments ..... **COVER**
- KFC 150 autopilot / flight director ..... **OFF (air-driven attitude gyro feeds the autopilot – expect disconnect)**
- Surface deice ..... **INOPERATIVE – exit icing immediately**
- IMC ..... **EXIT / AVOID – partial panel**
- Deferred: Vacuum / pneumatic failure – partial panel, no autopilot, boots inoperative; prefer VMC approach. ..... **NOTED**

Pitot / Static Source Blockage

Airspeed erratic or zero, altimeter / VSI frozen

Pitot heat ..... **ON**

Attitude and power ..... **FLY KNOWN pitch / power settings; GPS ground speed as cross-check**

▶ **Static system blocked?**

Yes:

- Cabin pressurisation ..... **DUMP / depressurise (below 10,000 ft) before using cabin static**
- Alternate static source ..... **ON (cabin) – only with cabin unpressurised; expect small altimeter / airspeed errors**
- Deferred: Static source blocked – alternate static in use, cabin depressurised, altimeter error on approach. ..... **NOTED**

No – pitot only:

- Airspeed indicator ..... **DISREGARD until heat clears the blockage; use attitude, power and GPS**
- Deferred: Airspeed unreliable – approach by attitude and power, add margin, longer runway. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff Roll

Throttle (memory) ..... **CLOSED**

Brakes (memory) ..... **AS REQUIRED – stop straight ahead**

Mixture ..... **IDLE CUTOFF**

Fuel selector ..... **OFF**

Magnetos ..... **OFF**

Battery master / alternator ..... **OFF**

Engine Failure After Takeoff

Airborne at low altitude

LAND IMMEDIATELY

Airspeed (memory) ..... **90 KIAS (best glide, gear & flaps up); 80–85 KIAS with gear / flaps down**

Fuel pump (memory) ..... **HI (if fuel flow low); fuel selector SWITCH TANKS if time permits**

▶ **Runway remaining to land on?**

Yes:

- Landing gear ..... **DOWN (if retracted)**
- Flaps ..... **AS REQUIRED**
- Land ..... **STRAIGHT AHEAD**

No – land ahead off-field:

- Landing gear ..... **DOWN on firm surface / UP on rough terrain or water**
- Flaps ..... **DOWN (36°) when landing assured**
- Cabin pressurisation ..... **DUMP (so the door opens after landing)**
- Fuel selector ..... **OFF**
- Mixture ..... **IDLE CUTOFF**
- Magnetos ..... **OFF**
- Battery master ..... **OFF (after gear position set)**

Land straight ahead with only small changes of direction. Do not attempt a turn back.

Engine Failure in Flight

Loss of power with altitude available

LAND AS SOON AS POSSIBLE

Airspeed (memory)	**90 KIAS (best glide, gear & flaps up)**
Fuel selector (memory)	**SWITCH TANKS (feel for detent)**
Fuel pump (memory)	**LO; HI if fuel flow reads zero (lean if engine floods)**
Mixture (memory)	**FULL RICH, then lean as required**
Propeller (memory)	**FULL AFT (low RPM) to extend glide**
Oxygen	**ON if cabin altitude climbs above 10,000 ft (pressurisation is lost without engine power)**

▶ Restart attempt practicable?

Yes – restart:

Magnetos	**BOTH – check**
Alternate induction air	**OPEN (if fitted / induction icing suspected)**
Throttle	**1/4 OPEN**
Starter	**ENGAGE if propeller has stopped**

▶ Engine restarted?

Yes:

Power	**AS REQUIRED – apply slowly**
Fuel pump	**LO if required**
Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel or induction system.	**NOTED**

No:

Continue with forced landing.

No – forced landing:

Forced landing without power:

ATC / ELT	**MAYDAY, squawk 7700**
Seats / belts / harnesses	**SECURE; passengers briefed**
Landing gear	**DOWN on firm surface (below 165 KIAS; use emergency extension if hydraulic pump fails) / UP on rough terrain or water**
Flaps	**DOWN when landing assured**
Airspeed	**80–85 KIAS on final**
Cabin pressurisation	**DUMP**
Fuel selector	**OFF**
Mixture	**IDLE CUTOFF**
Magnetos	**OFF**
Battery master / alternator	**OFF (after gear and flaps set)**
Cabin door	**UNLATCH before touchdown**

Engine Fire in Flight

LAND IMMEDIATELY

Fuel selector (memory)	**OFF**
Mixture (memory)	**IDLE CUTOFF**
Fuel pump (memory)	**OFF**
Throttle (memory)	**CLOSED**
Cabin pressurisation (memory)	**DUMP (bleed air OFF to cabin); cabin heat / defroster OFF**
Oxygen	**ON if smoke in cabin / cabin altitude above 10,000 ft**
Airspeed	**INCREASE (up to 165 KIAS) to help extinguish fire**
Landing gear	**DOWN while battery power remains (emergency extension if needed)**
Forced landing	**EXECUTE (do not attempt restart)**

Engine Fire During Start

Starter (memory) ..... CONTINUE CRANKING

▶ Engine starts?

Yes:

- Throttle ..... 1,700 RPM for a few minutes
- Engine ..... SHUT DOWN and inspect for damage

No:

- Mixture ..... IDLE CUTOFF
- Throttle ..... FULL OPEN
- Fuel pump ..... OFF
- Fuel selector ..... OFF
- Starter ..... CONTINUE CRANKING to draw fire into engine
- Magnetos ..... OFF
- Battery master / alternator ..... OFF
- Fire extinguisher ..... OBTAIN / USE
- Aircraft ..... EVACUATE

Electrical / Cabin Fire and Smoke

LAND AS SOON AS POSSIBLE

Battery master / alternator / standby alternator (memory) ..... OFF

Avionics master (memory) ..... OFF

Cabin heat / defroster / vents ..... OFF / CLOSED

Fire extinguisher ..... ACTIVATE (then ventilate cabin)

Oxygen ..... CREW MASKS ON if smoke persists (keep oxygen away from flames)

▶ Fire out and electrical power required?

Yes:

- Circuit breakers ..... CHECK – do not reset faulted
- Battery master ..... ON
- Alternator / avionics / equipment ..... ON one at a time, check for smoke
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... NOTED

No:

Landing gear hydraulic pump and flaps are electric: plan an emergency free-fall gear extension and a flaps-up landing.

- Deferred: Electrical power off – emergency gear extension, flaps-up landing, no landing light, no gear lights. ..... NOTED

Smoke evacuation:

Descent ..... TO 10,000 ft or below if practicable

Cabin pressurisation ..... DUMP (below 10,000 ft) to ventilate

Fresh air vents / storm window ..... OPEN

Emergency Descent

Throttle (memory) ..... CLOSED (reduce smoothly if time permits)

Propeller (memory) ..... FULL FORWARD

Landing gear (memory) ..... DOWN (below 165 KIAS)

Flaps (memory) ..... 10° (below 165 KIAS)

Airspeed (memory) ..... 165 KIAS (smooth air) / 135 KIAS (turbulence)

Oxygen ..... ON if cabin altitude above 10,000 ft

Mixture ..... ENRICH as required

ATC ..... ADVISE – emergency descent

Level-off ..... 10,000 ft or MEA – restore power slowly; gear up below 130 KIAS if continuing

Rapid Decompression / Cabin Altitude High

CABIN ALT warning (approx 10,000 ft cabin), ears / fog / loud decompression, or cabin climbing uncontrolled

LAND AS SOON AS PRACTICAL

Oxygen masks (crew) (memory)	**DON – oxygen ON, 100%**
Passenger oxygen (memory)	**ON – masks DON**
Emergency descent (memory)	**INITIATE to 12,500 ft or below (10,000 ft preferred)**
ATC	**ADVISE**
Pressurisation	**CHECK: dump switch NORMAL, controller set below flight altitude, manifold pressure above approx 20 in Hg, DOOR AJAR out**

▶ Cabin pressurisation recovered?

Yes:

Cabin altitude	**MONITOR; stay at an altitude the cabin holds below 10,000 ft**
Deferred: Pressurisation fault – low cruise altitude, oxygen used, maintenance inspection required.	**NOTED**

No:

Altitude	**REMAIN at or below 12,500 ft (10,000 ft without oxygen for passengers)**
Deferred: Unpressurised – remain low, oxygen quantity limited, no high-altitude diversion.	**NOTED**

Emergency Gear Extension (Free Fall)

Gear does not extend hydraulically / HYD PUMP light stays on

Airspeed	**REDUCE to approx 100 KIAS (below 165 KIAS Vle; slower aids free fall)**
Hydraulic pump circuit breaker	**PULL**
Landing gear selector	**DOWN**
Emergency gear extension lever	**PULL / EMERGENCY position (floor, between front seats) – hydraulic pressure released, gear free-falls**
Aircraft	**YAW left and right to assist gear into locks**
Landing gear indicator	**THREE GREEN**

Do not reset the hydraulic pump circuit breaker or move the gear selector up after an emergency extension. Leave the lever in the emergency position and the gear down.

▶ Three green obtained?

Yes:

Deferred: Gear extended by free fall – do not retract; gear and hydraulic system inspection required.	**NOTED**

No:

Continue to Gear-Up / Partial Gear Landing.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Passenger briefing	**COMPLETE (belts tight, brace, exits)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**DOWN (36°)**
Approach	**NORMAL, 80 KIAS**
Fuel pump	**OFF**
Cabin pressurisation	**DUMP**
Cabin door	**UNLATCHED**

On short final:

Mixture	**IDLE CUTOFF**
Fuel selector	**OFF**
Magnetos	**OFF**
Battery master / alternator	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped**

Inadvertent Icing Beyond Capability

Ice accumulating faster than the deice system clears it, or aircraft not approved for icing

LAND AS SOON AS PRACTICAL

Pitot heat / stall warning heat (memory)	**ON**
Propeller deice / windshield heat (memory)	**ON (if fitted)**
Surface deice (boots) (memory)	**CYCLE at first sign of ice – repeat as required (if fitted)**
Exit icing (memory)	**TURN BACK / CLIMB / DESCEND – ATC advised, request new altitude**
Power	**INCREASE to hold airspeed; manifold pressure monitored (induction ice – alternate air if fitted)**
Autopilot	**OFF – hand fly, note control feel and trim**
Airspeed	**MAINTAIN higher than normal; no abrupt pitch changes**
Cabin heat / defroster	**FULL ON – windshield hot plate for forward view**

▶ Residual ice on airframe for landing?

No:

Ice protection	**OFF when clear**

Yes:

Flaps	**10° maximum for landing**
Approach speed	**INCREASE 10–15 KIAS (approx 90–95 KIAS); expect stall warning unreliable**
Landing	**LONGER RUNWAY, minimal flare, no steep turns**
Deferred: Residual airframe ice – flaps 10°, approach 90–95 KIAS, longer landing distance.	**NOTED**

Spin Recovery

Intentional spins prohibited – inadvertent spin entry

Throttle (memory)	**IDLE**
Ailerons (memory)	**NEUTRAL**
Rudder (memory)	**FULL OPPOSITE to rotation**
Control wheel (memory)	**FULL FORWARD (release back pressure)**
Rudder (memory)	**NEUTRAL as rotation stops**
Recovery	**SMOOTH pull-out from dive; do not exceed 203 KIAS (Vne) or load limits**
Flaps / gear	**UP during recovery if extended**
Power	**RESTORE slowly once in level flight**

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist 7f510494 · 2026-11-07